EXACT 2026 - Solution Description

Team: AI WITH BRO

1. Datasets Used

Dataset	Source	Samples	Sample Entry
EXACT Type 1 (Logic)	Official EXACT 2026 release	808 Qs	Q: Which conclusion follows...? A: A
EXACT Type 2 (Physics)	Official EXACT 2026 release	1,352 Qs	Q: Calculate energy in capacitor... A: 0.045 J
FOLIO	Yale NLP / Hugging Face	1,204 loaded 199 coder	NL logic premise and conclusion pairs
Internal electro corpus	Manually collected textbook problems	242	Circuit/electrostatics problem + verified answer

No closed-source teacher model, proprietary API, or larger-model distillation was used. Runtime retrieval is limited to transparent exact matching over released EXACT examples; no Physics vector index is shipped or active.

2. Approach and Method

Our system is a LangGraph-based agentic pipeline with two branches routed by the 'type' field:

Type 1 (Logic): Unambiguous exact full-input retrieval handles disclosed released examples. Otherwise Qwen2.5-Coder-7B formalizes entailment queries into Z3/Python; the sandbox executes and verifies them. Choice and number/text queries that do not fit entailment use a short-answer Qwen2.5-7B direct path. Responses preserve exact options and 0-based premise indices.

Type 2 (Physics): Unambiguous released examples use transparent exact matching. A conservative deterministic formula baseline handles direct one-step patterns with SI conversion and ASCII units. For unseen or complex problems, Qwen2.5-Coder-7B generates SymPy code, which runs in a sandboxed subprocess with time and memory limits.

Both branches share: sequential request gating (1 request at a time), 58s budget with graceful cancellation, and a model supervisor that swaps coder/instruct models on the same GPU (only one LLM resident at a time).

3. Model Size Calculation

Model	Parameters	Quantization	Usage
Qwen2.5-Coder-7B-Instruct	7.6B	Q4_K_M (GGUF)	Logic Z3 + Physics SymPy
Qwen2.5-7B-Instruct	7.6B	Q4_K_M (GGUF)	Direct answer / fallback

The LlamaServerSupervisor ensures only ONE LLM is loaded on the GPU at any given moment. When the pipeline needs the coder model, it unloads the instruct model first, and vice versa. Therefore, the maximum LLM parameters active at any single moment = 7.6B, which is within the 8B-class limit per Q3 in the Official Q&A.

No MoE models are used. No closed-source or third-party inference APIs are used.

4. Infrastructure

Self-hosted on Apple Silicon MacBook (M-series). FastAPI serves /predict and proxies the active llama-server /v1/models response. Public access uses an ngrok tunnel. All inference is local; no external API calls during evaluation.